Code No: 135CE **R16**

JAWAHARLAL NEHRU TECHNOLOGICAL UNIVERSITY HYDERABAD
B. Tech III Year I Semester Examinations, March - 2021
ENVIRONMENTAL ENGINEERING
(Common to CE, EEE, CSE, IT, MIE)

Time: 3 Hours **Max. Marks: 75**

Answer any five questions
All questions carry equal marks
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USED PAPER MARCH-2021

1. The population data of a city is given as under:

Year	1961	1971	1981	1991	2001	2011
Population	76000	90000	115000	150000	187000	225000

Estimate the population of the city in 2021 and 2025 using any four standard methods. [15]

2. Explain the drinking water standards of physical, chemical, bacteriological and radiological characteristics. [15]

3. Elaborate a slow sand filter construction. In what way it is different from a rapid sand filter? [15]

4. Enumerate the design criteria for sedimentation tank and clariflocculator. [15]

5. A district, having an area of 16 hectares with coefficient of relative impermeability as 0.75, has to be drained by a sewer of 900 metres length, laid at a gradient of 1 in 300. The rainfall intensity may be assumed to be of 10 to 20 minutes duration and the time of entry at the sewer inlet as 3 minutes. Design the storm water sewer. [15]

6. Compare conservancy and water carriage systems of sanitation indicating the merits and demerits of each. [15]

7. Under what circumstances is pumping of sewage necessary? Which type of plumbing system of drainage most commonly used in India? Discuss essential safeguards need to be provided. [15]

8. A channel type grit chamber has a flow through velocity of 0.25 m/s, a depth of 0.8 m and a length of 10m for inorganic grit particles with specific gravity 2.5, determine the particle size that can be removed with 100% efficiency. (Assume $v=1.01\times10^{-2}$ cm^2/s). [15]

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